Marcel Bruckner

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marcel-bruckner • MarcelBruckner

Skills

category 1: XXX, YYY, ZZZ
category 2: XXX, YYY, ZZZ
category 4: XYZ
category 5: XYZ
category 4: XXX, YYY, ZZZ
category 5: XXX, YYY, ZZZ
category 6: XYZ

Education

TUM School of Computation, Information and Technology **TU Munich**
Informatics: Games Engineering – M.Sc., Final grade: 1.6 *Apr. 2019 – Mar. 2022*
My specialization is in computer vision, machine learning, deep learning, and robotics. I conducted practicals and research projects on autonomous driving and robotic applications, applying my knowledge and skills.

TUM School of Computation, Information and Technology **TU Munich**
Informatics: Games Engineering – B.Sc., Final grade: 2.1 *Oct. 2015 – Mar. 2019*
I gained a comprehensive understanding of computer science fundamentals, including software engineering, algorithms, mathematics, databases, and distributed systems. I specialized in computer graphics, physics simulations, augmented/virtual reality, AI, and autonomous driving.

Master thesis

Vision-Based Continual Reinforcement Learning for Robotic Manipulation Tasks [1] **TU Munich**
TUM: Chair of Robotics, Artificial Intelligence, and Real-time Systems, Final grade: 1.0 *Oct. 2021 – Mar. 2022*
I merged techniques from continual learning and reinforcement learning on a single arm robot in a monocular camera setting and developed a hypernetwork-based approach to train the robot using only observations from a single camera.

Experience

CQSE GmbH **Munich**
Software Consultant *Mar. 2022 – Today*
I advise our customers in establishing effective software quality assurance processes by analyzing their software systems, architecture, used technologies, and their development process, all with a focus on future-proofness and maintainability.

TUM: Chair of Robotics, Artificial Intelligence and Real-time Systems **Munich**
Student Assistant with a Degree *Apr. 2021 – Sep. 2021*
I implemented the Online Self-Correcting Calibration Architecture for Multi-Camera Traffic Localization Infrastructure for the Providentia++ project on the A9 highway. We documented our work in the following publication [2].

Publications

An Online Self-Correcting Calibration Architecture for Multi-Camera Traffic Localization Infra. [2] **IEEE**
IEEE Symposium on Intelligent Vehicle *2024*
Most vision-based sensing and localization infrastructure today employ conventional area scanning cameras due to the high information density and cost efficiency offered by them. While the information-rich two-dimensional images provided by such sensors make it easier to detect and classify traffic objects with the help of deep neural networks, their accurate localization in the three-dimensional real world also calls for a reliable calibration methodology, that maintains accuracy not just during installation, but also under continuous operation over time. In this paper, we propose a camera calibration architecture that extracts and uses corresponding targets from high definition maps, augment it with an efficient stabilization mechanism in order to compensate for the errors arising out of fast transient vibrations and slow orientational drifts. Finally, we evaluate its performance on a real-world test site and showed to improve the localization accuracy by 50%.

Interests

Selfhosting: Homelab for personal services, smart home and home surveillance system

Sports: Gym, Disc Golf, Downhill–Biking

Volunteering: Passionate Blood–Donor, Foster home for saved stray dogs coming from greece

Languages

German: Native
English: Fluent

References

[1] Marcel Bruckner. Vision-based continual reinforcement learning for robotic manipulation tasks. `https://github.com/MarcelBruckner/CDN/blob/main/master-thesis.pdf`, 15.02.2022.

[2] Leah Strand, Marcel Bruckner, Venkatnarayanan Lakshminarasimhan, and Alois Knoll. An online self-correcting calibration architecture for multi-camera traffic localization infrastructure. In *2024 IEEE Intelligent Vehicles Symposium (IV)*, pages 1666–1671, 2024.

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Company Recruitment team
Company, Inc.
123 somestreet
some city

March 8, 2025

Job application

Dear Sir or Madam,

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis ullamcorper neque sit amet lectus facilisis sed luctus nisl iaculis. Vivamus at neque arcu, sed tempor quam. Curabitur pharetra tincidunt tincidunt. Morbi volutpat feugiat mauris, quis tempor neque vehicula volutpat. Duis tristique justo vel massa fermentum accumsan. Mauris ante elit, feugiat vestibulum tempor eget, eleifend ac ipsum. Donec scelerisque lobortis ipsum eu vestibulum. Pellentesque vel massa at felis accumsan rhoncus.

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Albert Einstein discovered that $e = mc^2$ in 1905.

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$$

Yours faithfully,

John Doe
Marcel Bruckner

Attached: curriculum vitæ